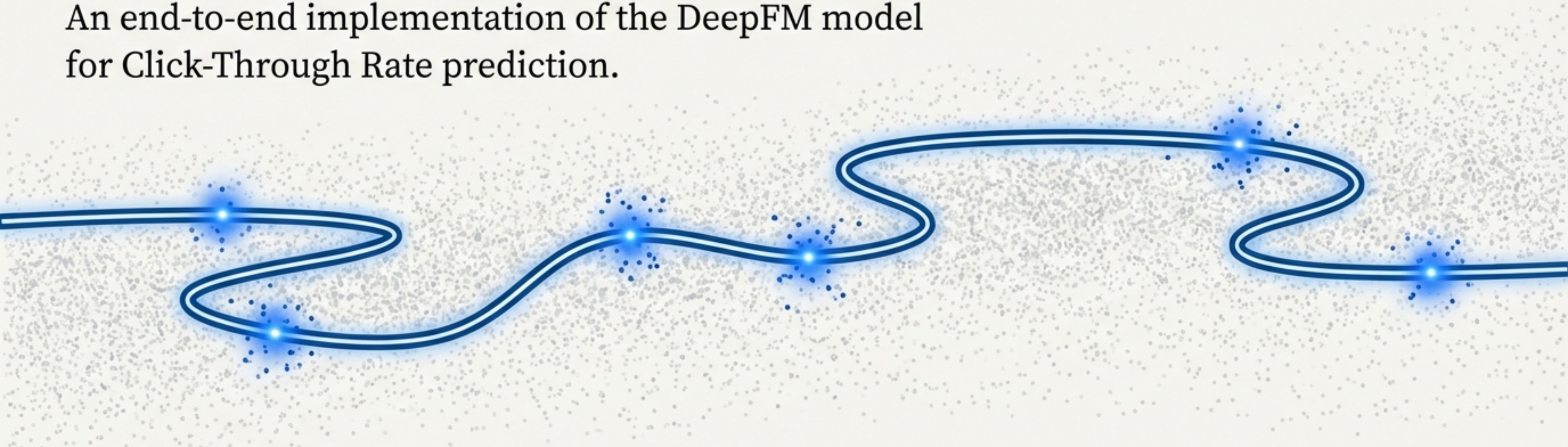


Predicting the Next Click

Source Serif Pro

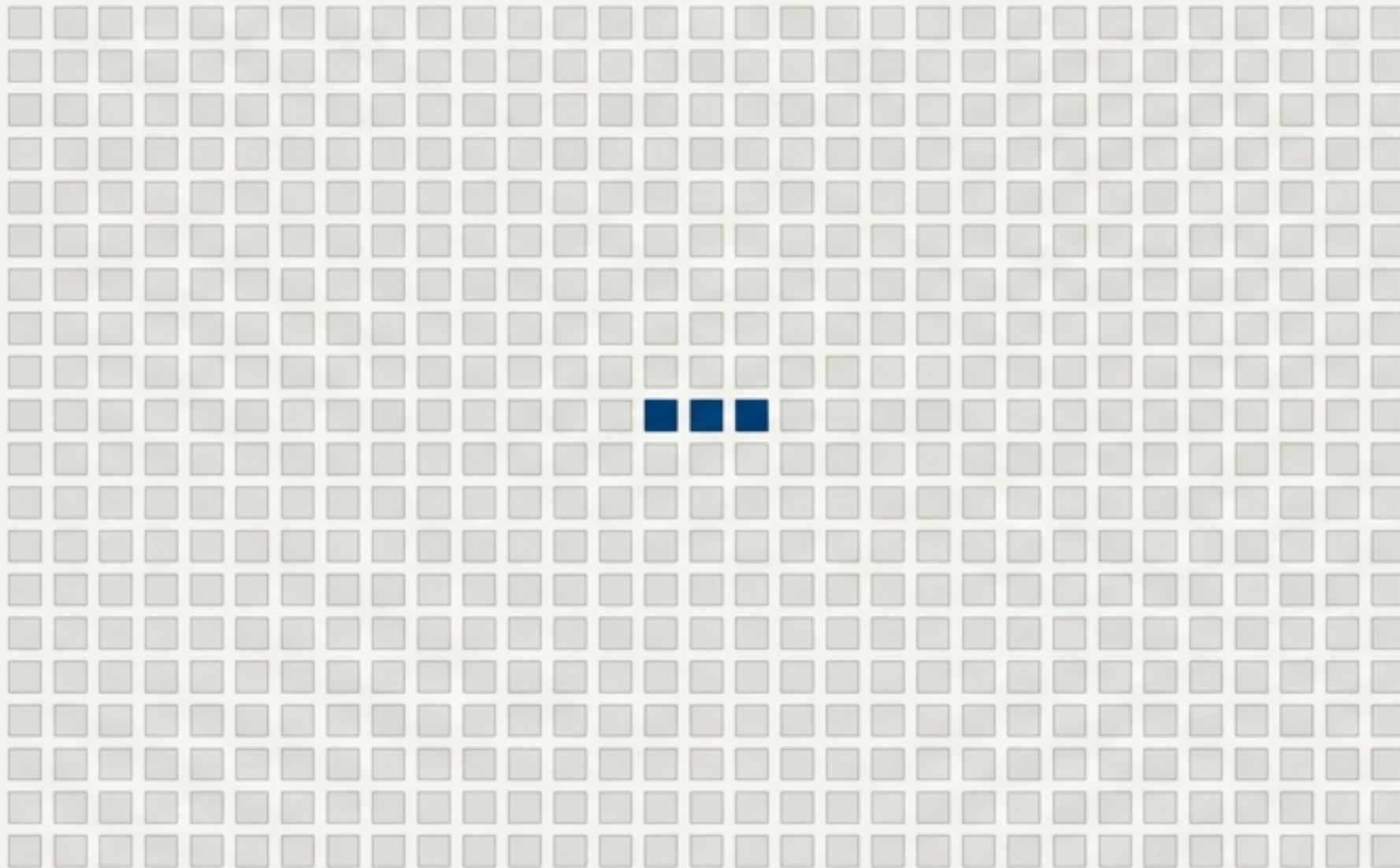
An end-to-end implementation of the DeepFM model
for Click-Through Rate prediction.



**Every second, millions of ads are shown online.
Which ones will be clicked?**

Click-Through Rate is the engine of the digital ad economy.

$$\text{CTR} = \frac{\text{Number of Clicks}}{\text{Number of Impressions}} \times 100\%$$



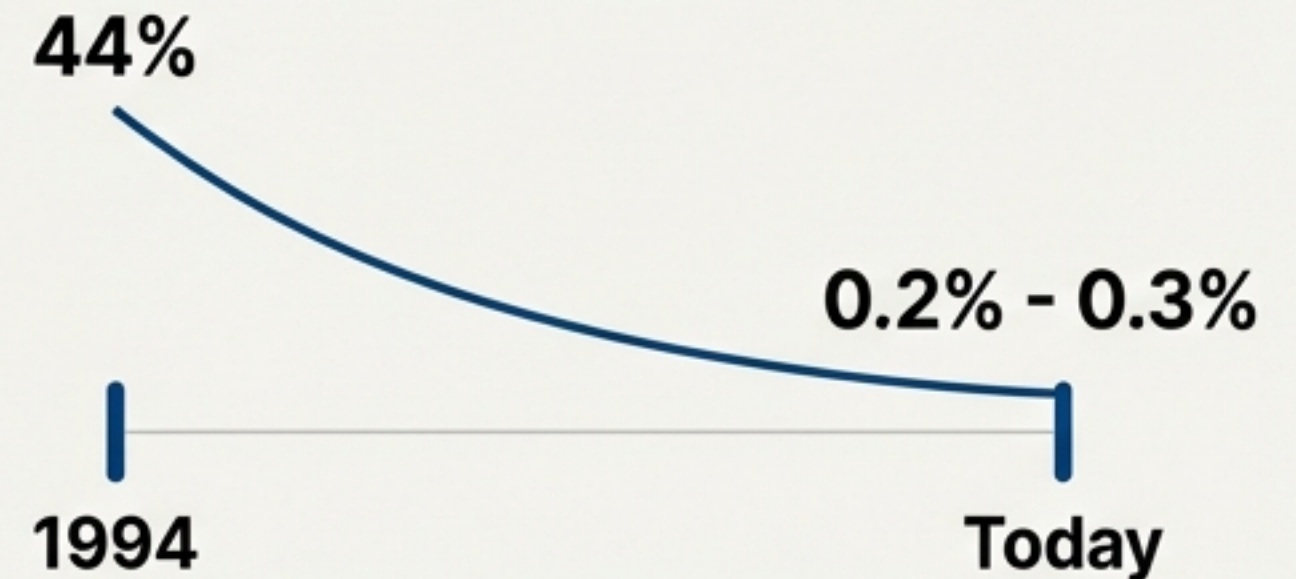
What is CTR?

The ratio of clicks on an ad to the number of times it is shown (impressions).

Why does it matter?

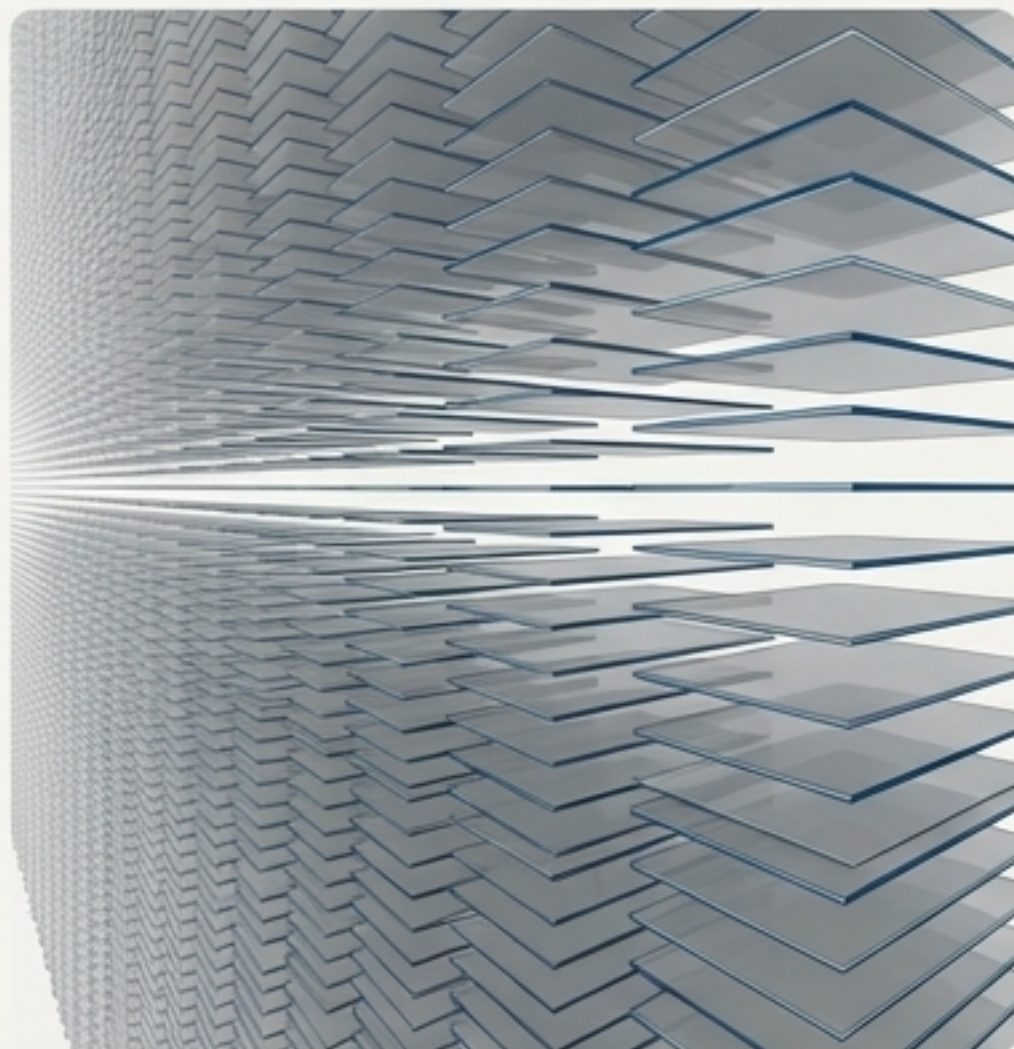
Accurate prediction is essential for:

- Effective ad ranking
- Revenue optimization
- Personalized user experience



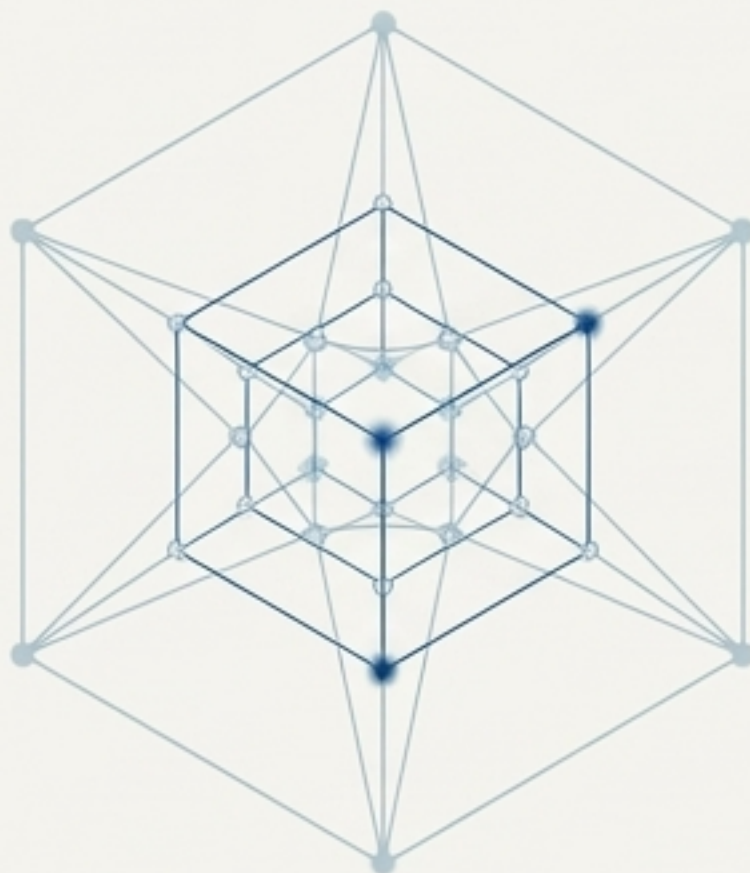
The data is the biggest challenge.

MASSIVE SCALE



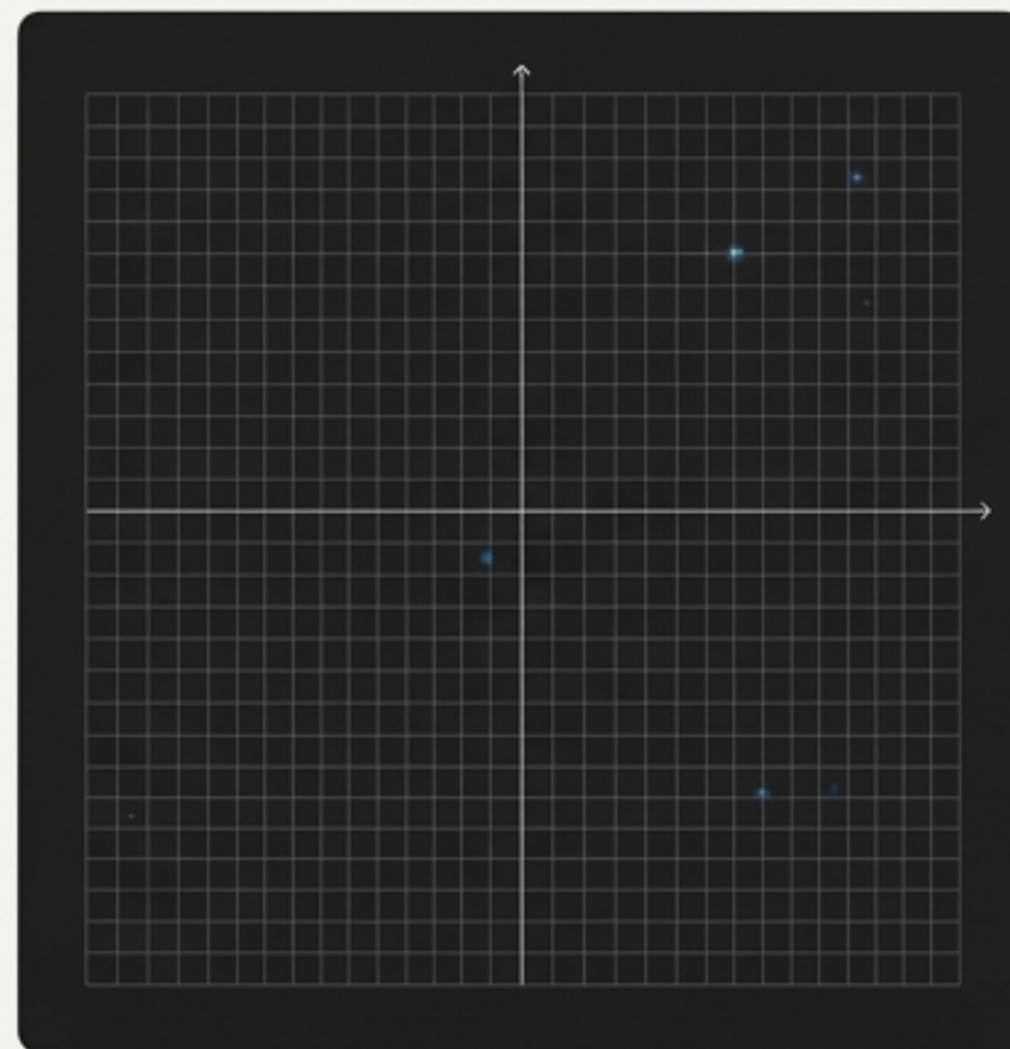
Datasets contain billions of samples with millions of unique features.

HIGH DIMENSIONALITY



User, item, and context features create an enormous feature space.

EXTREME SPARSITY



When one-hot encoded, this results in input vectors where almost all values are zero, making it difficult to find meaningful patterns.

The Proving Ground: The Criteo Dataset

SOURCE

A widely-used public benchmark from Criteo's Display Advertising Challenge.

SCALE

**~46
million**

records of ad impressions and click feedback.

FEATURES

A classic mix of:

- **13 Dense Features** (e.g., counts)
- **26 Sparse Features** (e.g., device type, ad category)

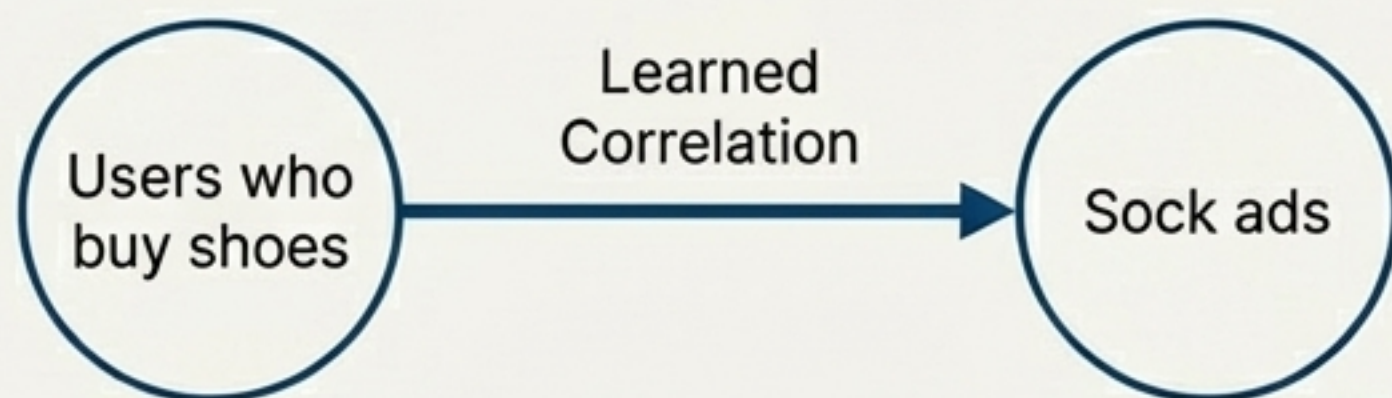
THE TASK

A binary classification problem—predict if an ad was clicked (1) or not (0).

DeepFM: A Model That Learns Both Simple and Complex Patterns

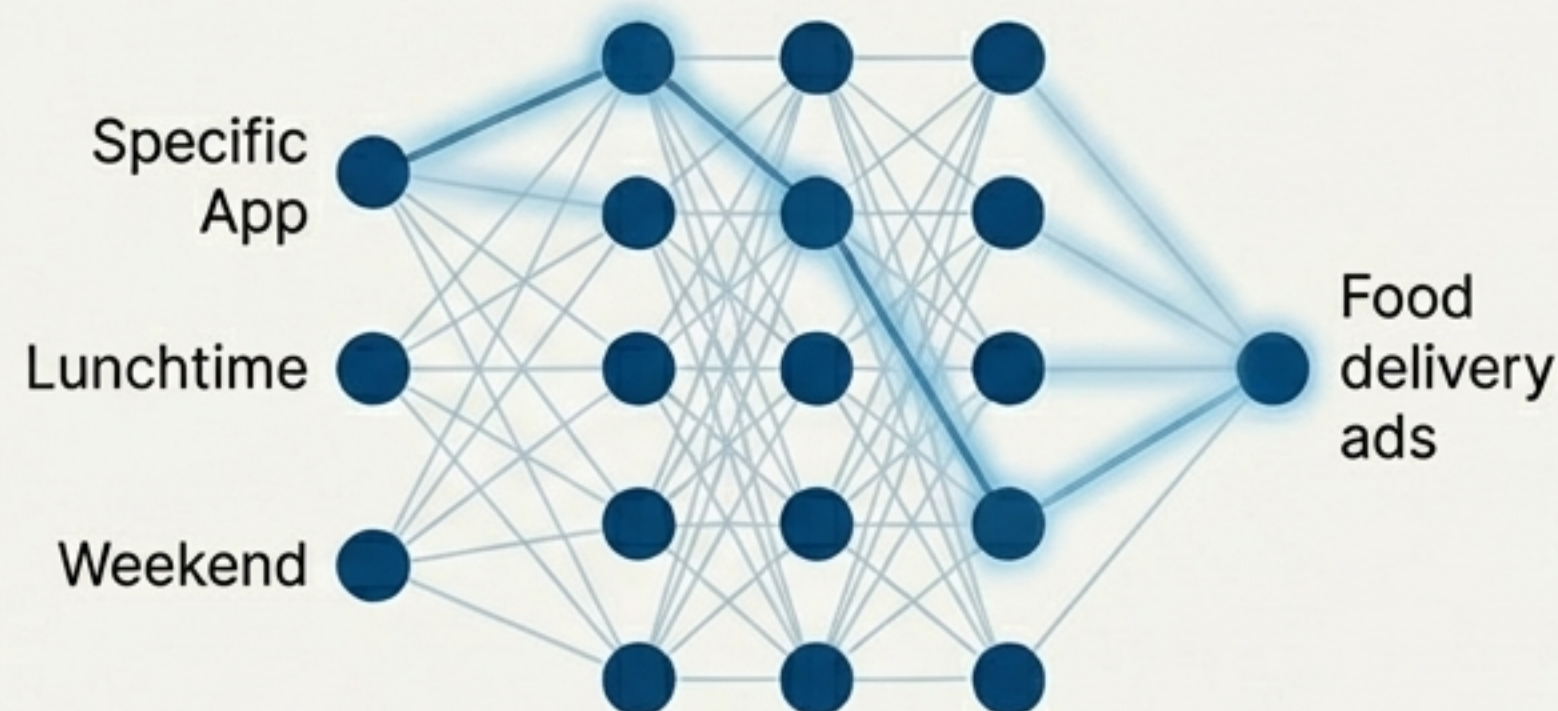
An end-to-end model combining Factorization Machines (FM) and Deep Neural Networks (DNN) without manual feature engineering.

Memorization FM Component ("Wide")



Excels at learning direct, low-order feature interactions.

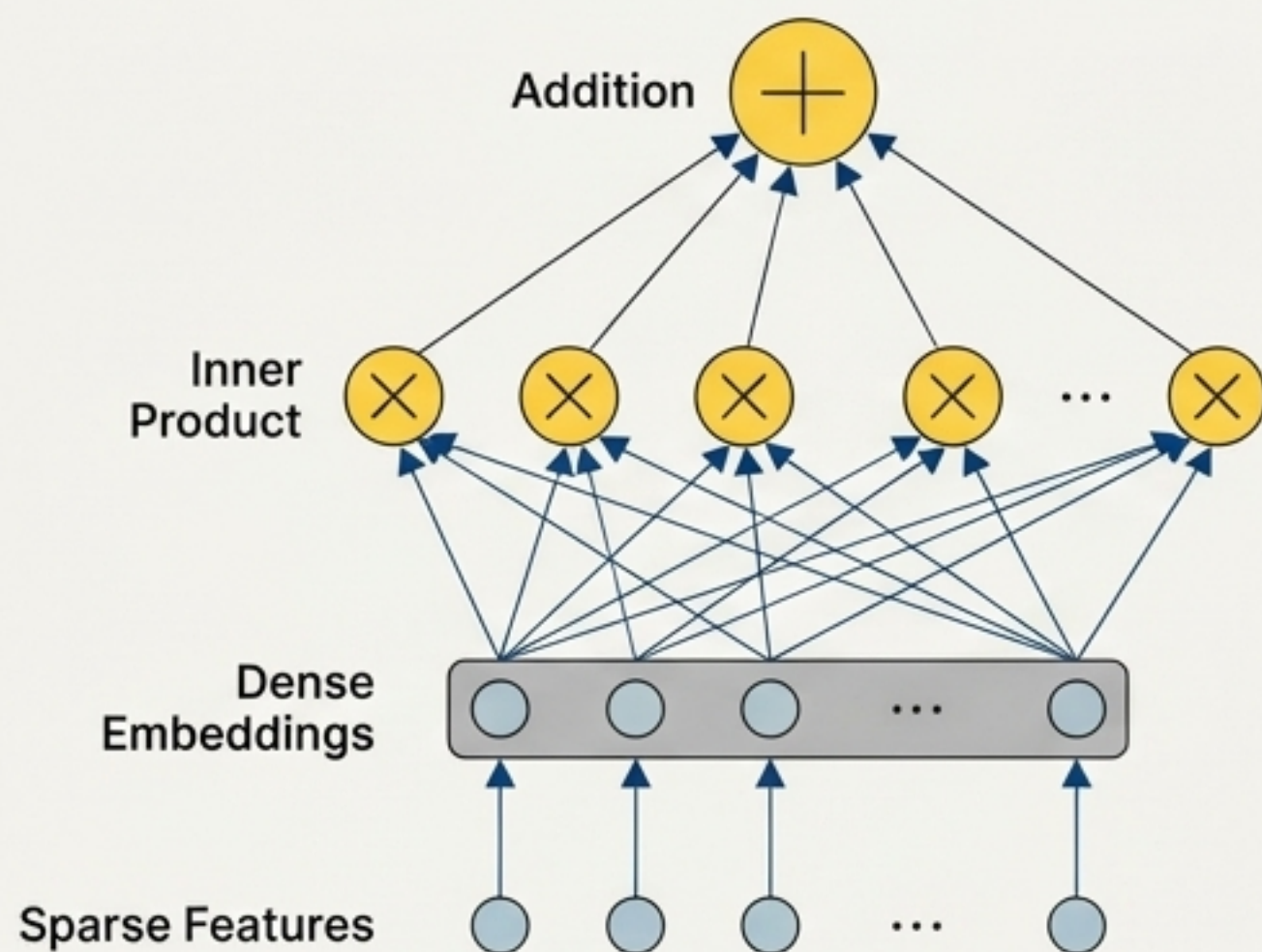
Generalization DNN Component ("Deep")



Discovers subtle, high-order patterns that are not obvious.

A Tale of Two Components

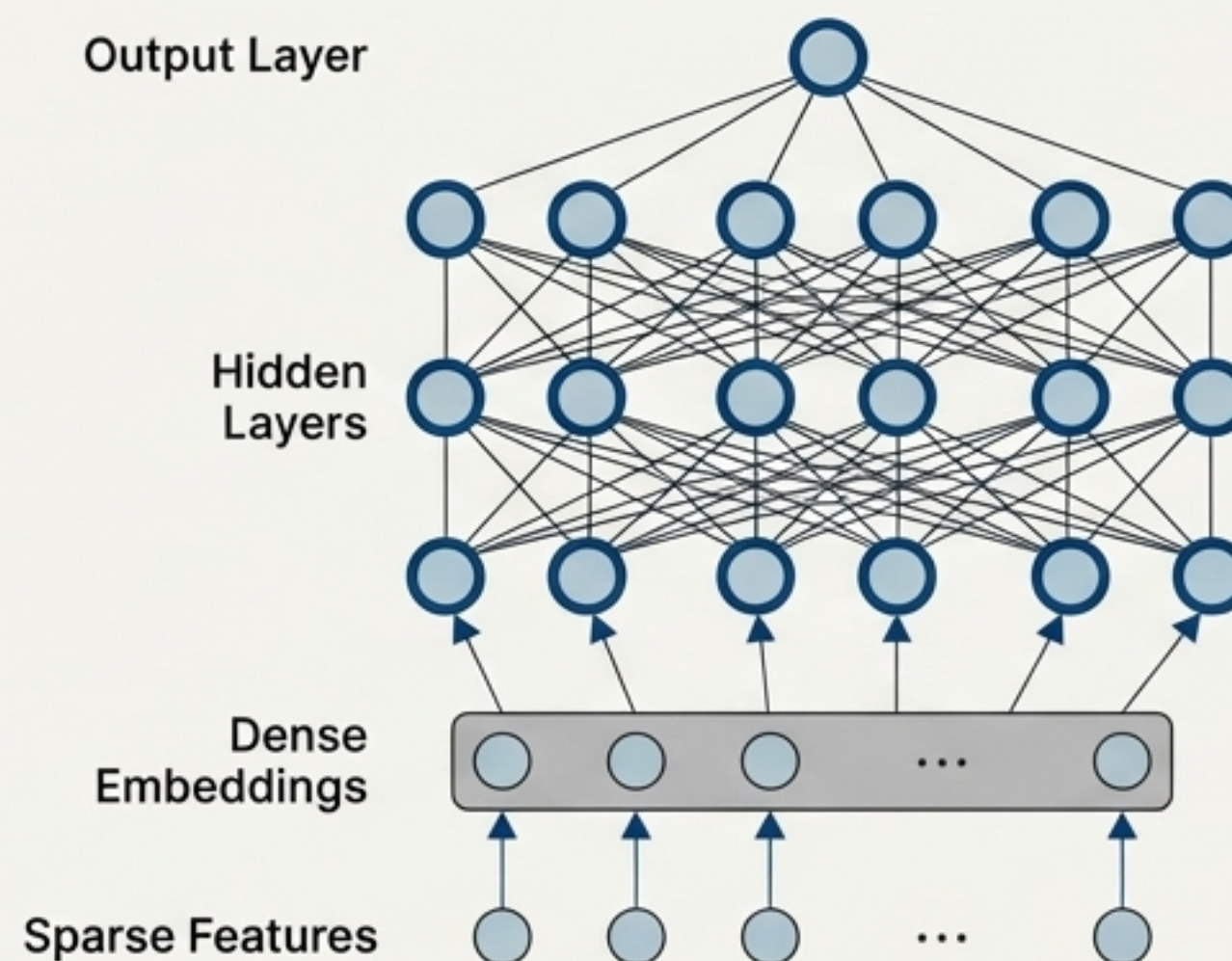
The Factorization Machine (FM) Layer



Purpose: To model pairwise (2nd-order) feature interactions.

Mechanism: Calculates interaction strength as the inner product of latent feature vectors. Highly effective for sparse data.

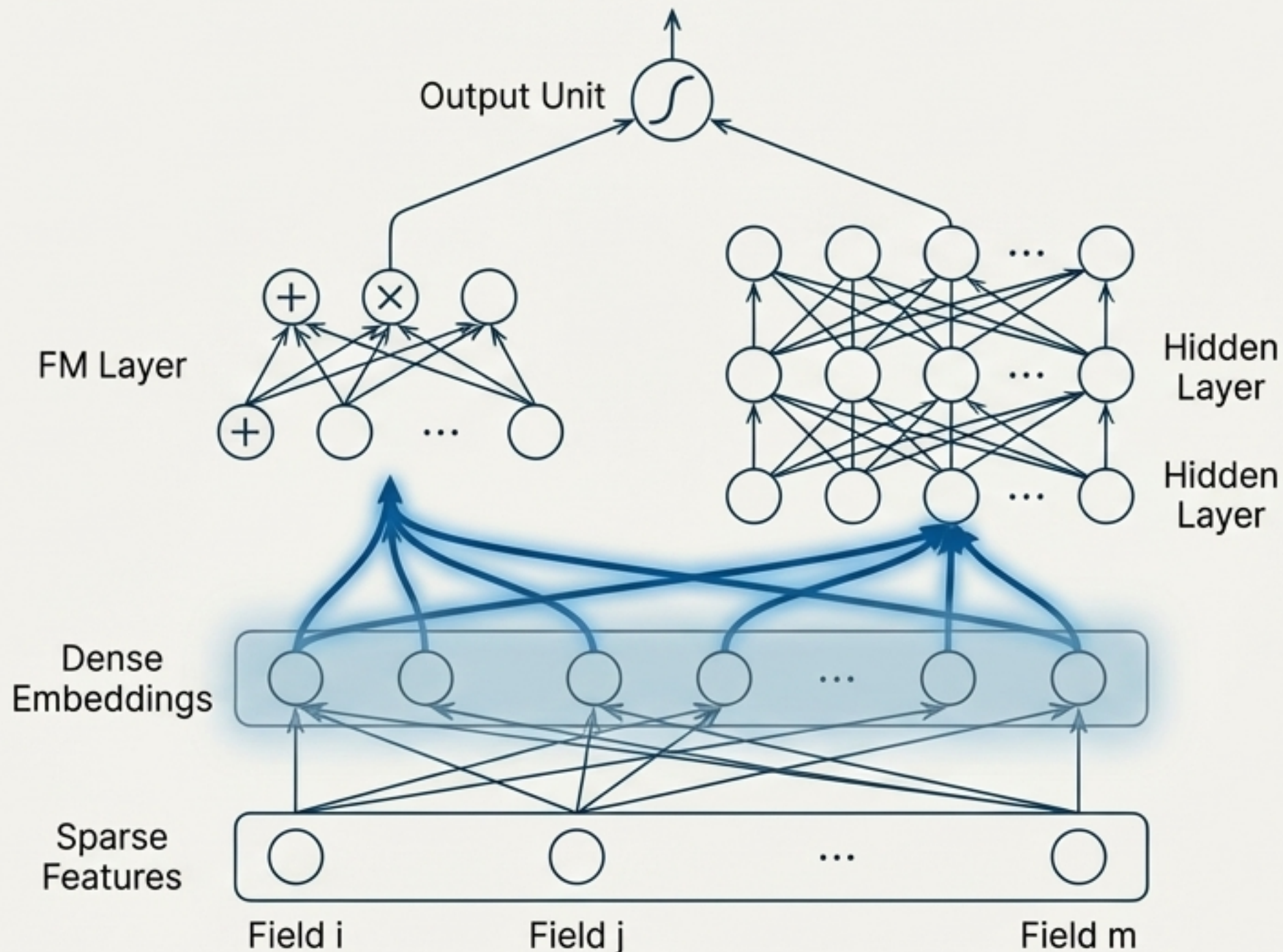
The Deep Neural Network (DNN)



Purpose: To learn high-order, complex feature interactions.

Mechanism: A feed-forward neural network that applies non-linear transformations to automatically discover hidden patterns.

The Key Innovation: A Shared Feature Embedding



****Why it Works****

- 1. Learns from Raw Features:** The shared embedding allows the model to learn low- and high-order interactions simultaneously.
- 2. No Feature Engineering:** Eliminates the need for manual, time-consuming feature crafting.
- 3. Efficient & Precise:** Embeddings are refined by gradients from both components, creating a richer feature representation.

Validating the Implementation

My results on the Criteo test set, mirroring the original DeepFM paper.

AUC (Area Under ROC Curve)

0.8007



Measures the model's ability to correctly rank positive examples higher than negative ones. Higher is better.

Logloss (Cross-Entropy)

0.4508

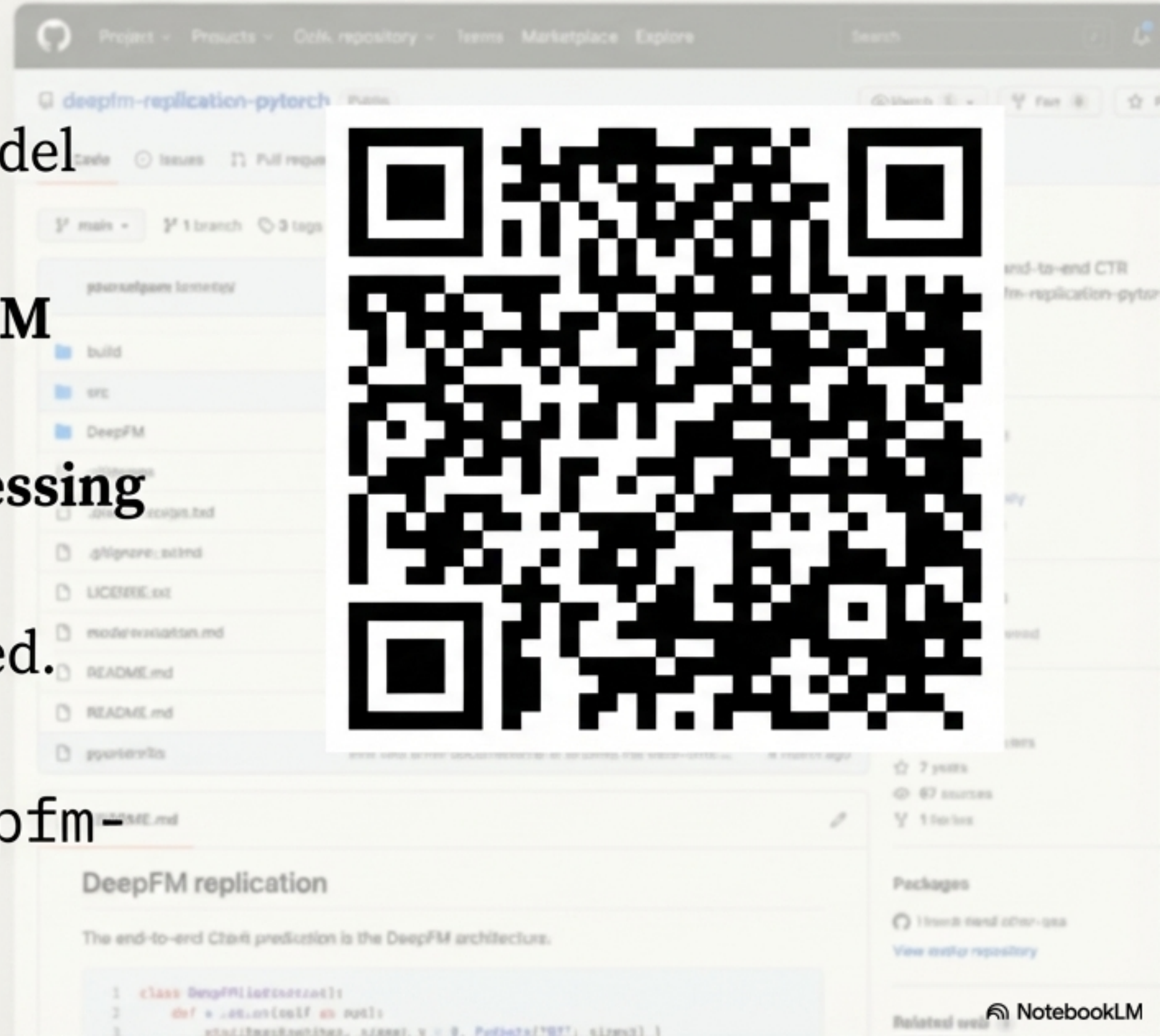


Penalizes the model for being confident in wrong predictions. Lower is better.

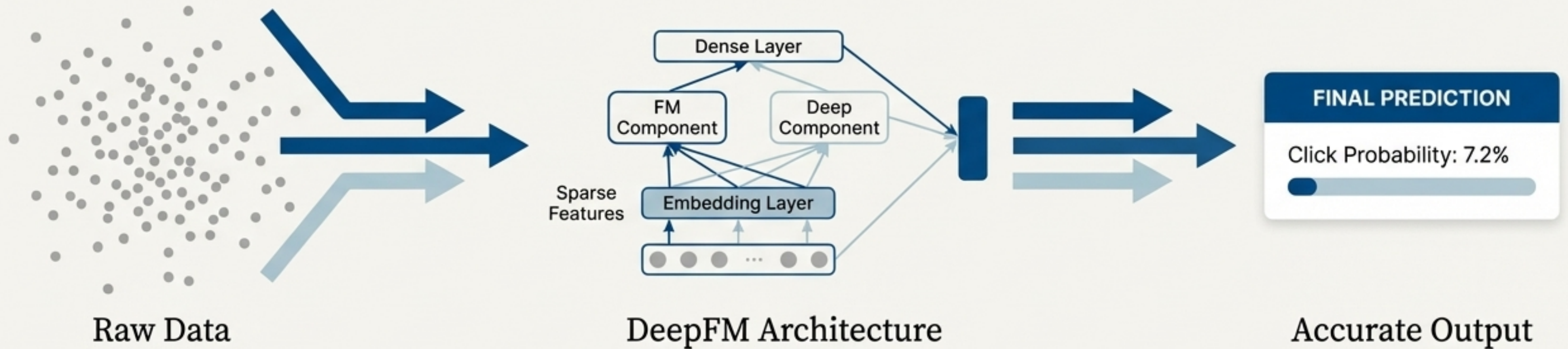
Explore the Code on GitHub

- An end-to-end CTR prediction model implemented in **PyTorch**.
- A faithful replication of the **DeepFM** architecture.
- Includes scripts for **data preprocessing** and **model evaluation**.
- Fully open-source and documented.

github.com/YourUsername/deepfm-replication-pytorch



From Raw Data to Accurate Predictions



The Challenge is Sparsity: CTR prediction is defined by finding signals in massive, sparse, and high-dimensional data.

Architecture is the Solution: DeepFM's hybrid design provides an elegant solution that bypasses manual feature engineering.

This Project Delivers: A complete, open-source PyTorch implementation for building upon this state-of-the-art model.